Futureproof

Equity Preservation Mortgage v14c

Model Review

Principal & Interest + Index-linked ETF Analysis

April 2025

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Executive Summary

This report reviews the FutureProof Equity Preservation Mortgage (EPM) v14c model as a piece of quantitative engineering — the stochastic processes, simulation mechanics, insurance pricing, and the Payments Waterfall. Its purpose is to document how the model works and to verify that the logic is internally consistent. A separate companion paper — "Model Assumptions and Parameter Risk" (April 2025) — examines whether the *input parameters* are empirically defensible.

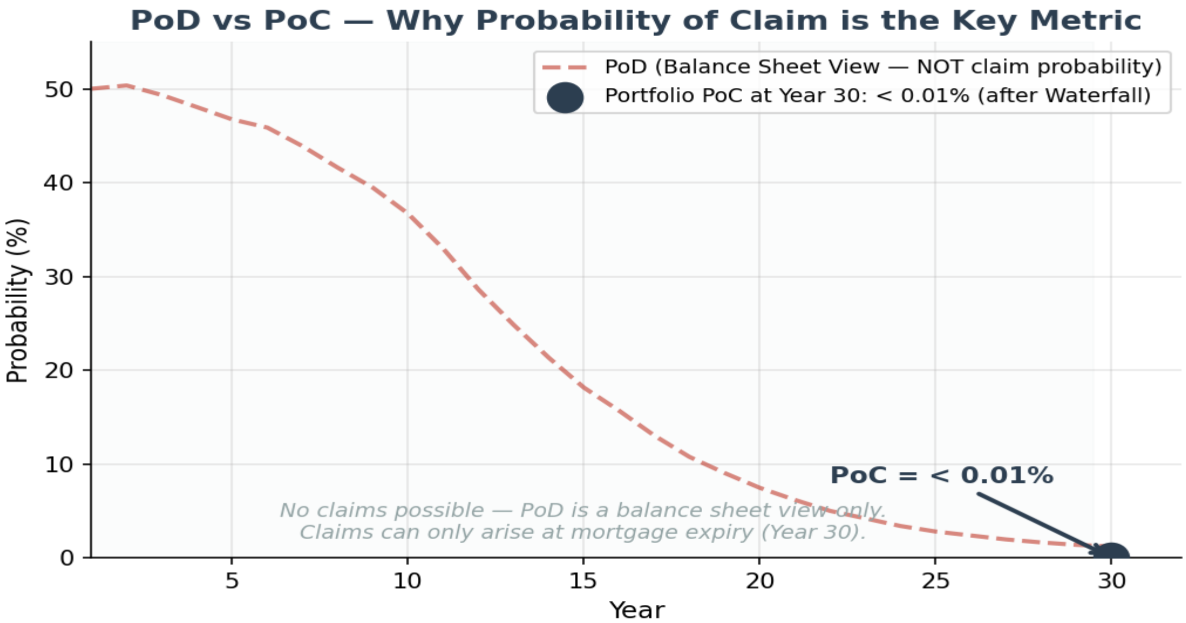
A critical distinction in this review is between Probability of Deficit (PoD) and Probability of Claim (PoC). PoD is a balance sheet snapshot — it shows what fraction of paths are in deficit at a given point in time. PoC is the actuarially relevant metric — it measures the probability of an actual insurance claim, which can only occur when a mortgage expires. PoC is the primary metric for insurance and reinsurance pricing.

All headline numbers in this report are *model-as-written* — computed at the base-case parameters (μ = 9.2%, γ = 0.163, θ = 2.13%, collar cost 0.046%). The companion Model Assumptions paper shows that under an empirically-centred parameter set, the base-case 1.2% PoD rises to 16.9%, and under an adverse-plausible set it rises to 43.2%. Users of this report should read the two documents together.

Key Findings

- Independent 50,000-path Monte Carlo confirms 1.2% PoD at base-case parameters (SE: 0.04%)
- **Portfolio PoC at year 30 is < 0.01% at base-case parameters** — after the Payments Waterfall is applied (ETF selldown, cross-subsidisation from surplus loans). The parameter-uncertainty range is discussed in the companion paper.
- Insurance claims can only arise at mortgage expiry (Year 30) — PoD before expiry is irrelevant to insurance
- Volatility buffer (cap 1.4 / floor 0.8) manages downside exposure on 16.6% equity vol
- Stochastic OU cash rates with equity correlation of 0.30 (appropriate for 30-year horizons)
- Insurance premium based on discounted expected deficit (S3), discounted at 2.13% (cash rate mean)
- LMI covers 80% of loss; worst 20% of deficit paths (larger losses) transferred as tail risk to reinsurance
- Surplus at maturity split 50/50 between FutureProof and Mortgage Funder
- Profit realisation every 5 years: 10% of surplus drawn every 5 years

PoD vs PoC — The Critical Distinction



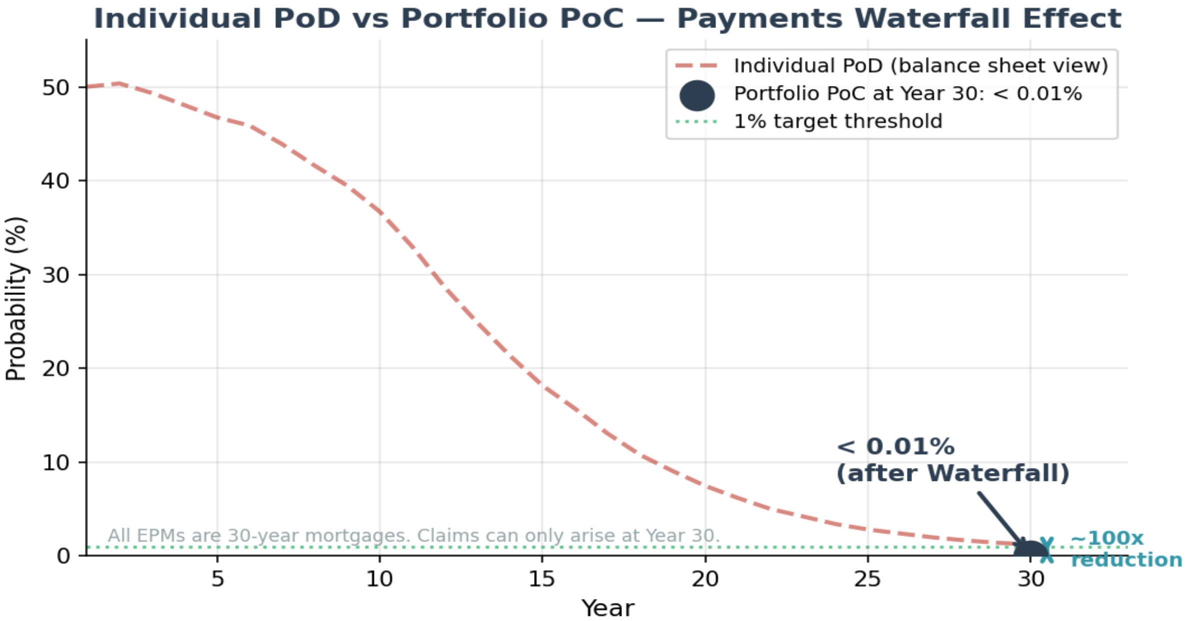
Probability of Deficit (PoD) is a balance sheet view — it shows whether the investment account is below the mortgage balance at a particular point in time. PoD is naturally high in early years (50.0% at year 1) because upfront costs and annuity payments create an immediate cost burden before investment compounding takes effect. It declines to 1.2% by year 30 as P&I; amortisation reduces the mortgage balance.

Probability of Claim (PoC) is the actuarially relevant metric. An insurance claim can **only** be made when a mortgage loan contract expires — any PoD prior to that point is irrelevant to insurance and reinsurance. This is why PoC is calculated only at mortgage expiry (Year 30). Furthermore, before any insurance claim is triggered, the **Payments Waterfall** is applied:

- **Step 1:** Sell down the ETFs in the expiring mortgage's investment account
- **Step 2:** Cross-subsidise by taking surpluses from other open mortgages in the portfolio
- **Step 3:** Only after Steps 1 and 2 is the net deficit known — this determines the actual PoC

This is why portfolio PoC (< 0.01% at year 30) is dramatically lower than individual PoD (1.2%). The Payments Waterfall is cooked into the portfolio structure and is the primary risk mitigation mechanism.

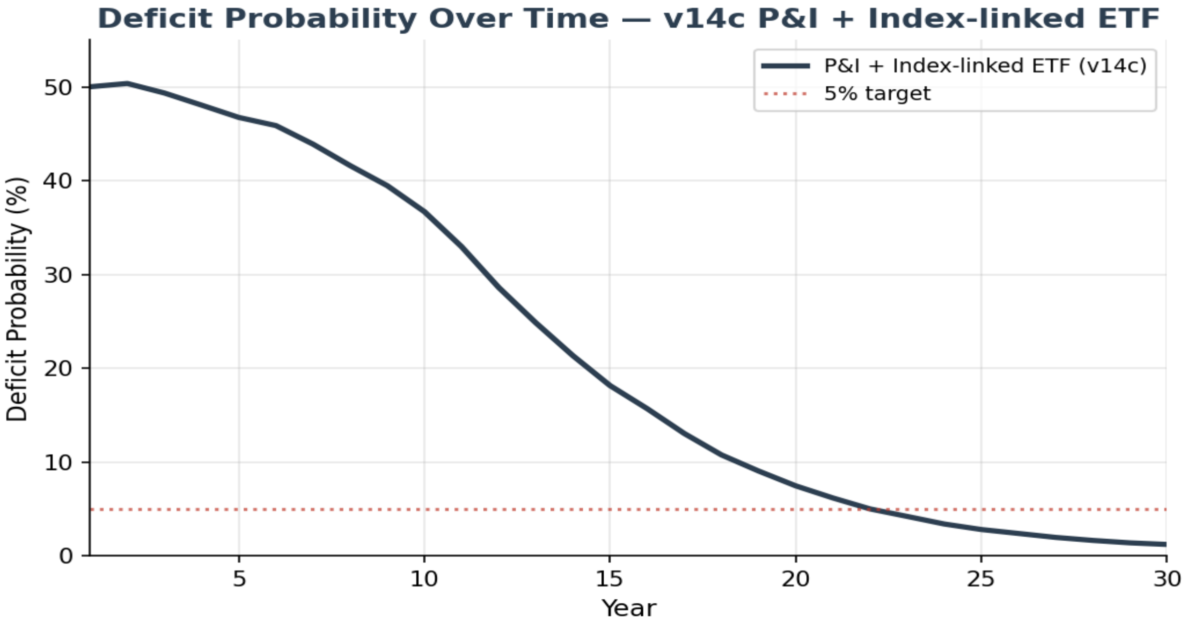
Portfolio Probability of Claim (PoC)



The chart above illustrates the critical distinction between PoD and PoC. The dashed red line shows individual PoD (balance sheet view) declining from 50.0% in Year 1 to 1.2% at Year 30 as P&I; amortisation takes effect. The single blue point shows portfolio PoC at Year 30 (the only point where claims can arise) after the Payments Waterfall — just < 0.01%. This ~100x reduction from PoD to PoC reflects the powerful effect of cross-subsidisation from surplus mortgages in a diversified portfolio.

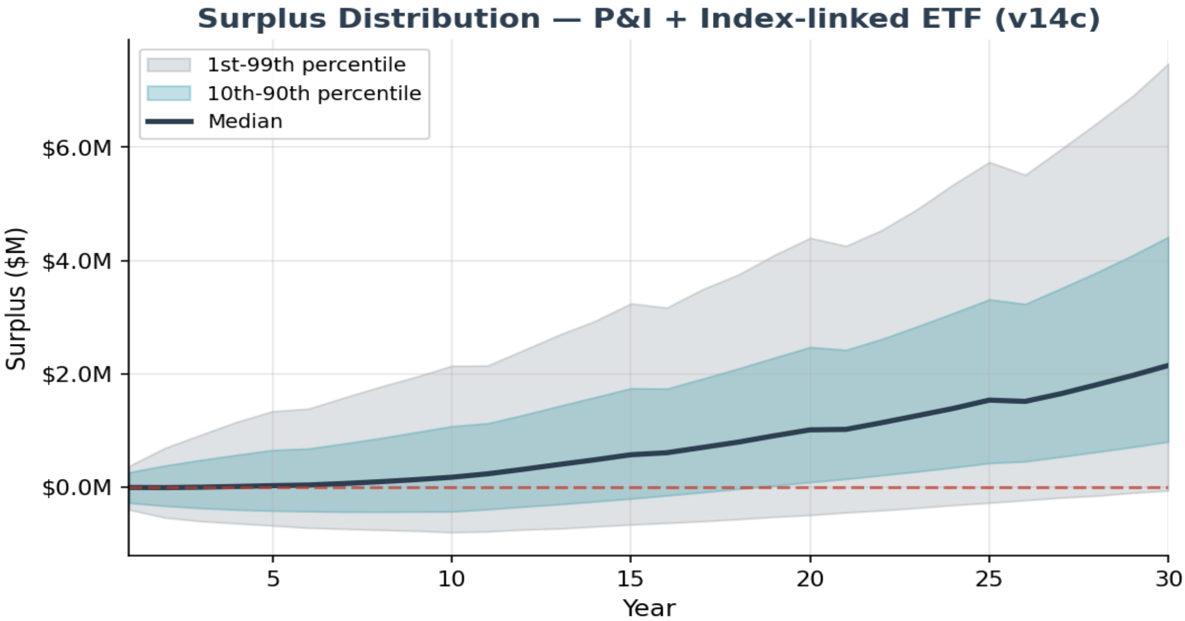
At < 0.01% PoC at year 30, the reinsurance claim probability is well below the 1% threshold that reinsurers typically require for acceptable risk transfer. This validates the commercial viability of the insurance structure.

Deficit Probability (PoD) Over Time — Balance Sheet View



PoD shows the fraction of paths in deficit at each point in time — a useful balance sheet view but **not** the probability of an insurance claim. PoD starts at 50.0% in year 1 and declines to 1.2% by year 30. The rapid decline after year 10 reflects P&I; amortisation — as the loan shrinks, the investment needs only modest returns to exceed the declining balance. Note: PoD at any point before mortgage expiry is irrelevant to insurance and reinsurance — only PoC on expiring mortgages matters.

Surplus Distribution — Fan Chart



The fan chart shows the distribution of surplus (investment balance minus mortgage balance) over the 30-year tenure. The median surplus reaches $2,151,975 at maturity. The 10th-90th percentile band widens over time but remains predominantly positive after year 15.

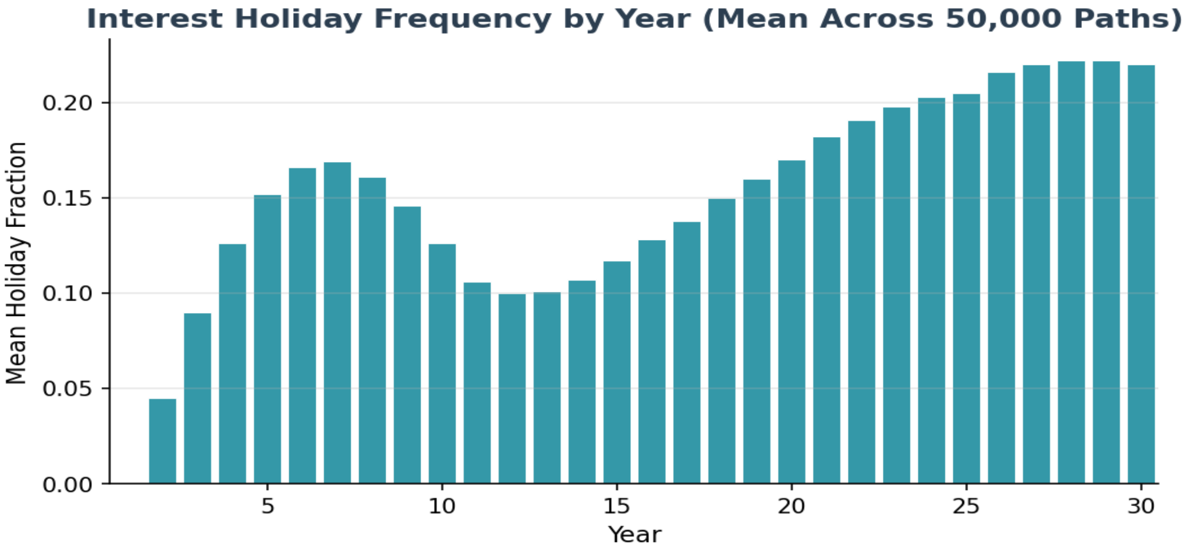
Important: Surplus at maturity is split 50/50 between FutureProof and the Mortgage Funder — it does not go to the borrower. Additionally, partial profit realisations occur every 5 years (10% of surplusdrawn). These periodic realisations are factored into the model.

Holiday Mechanism Analysis

The interest holiday mechanism is a protective feature that temporarily pauses interest charges when the investment account falls below a threshold (0.9x the initial loan). The 50,000-path Monte Carlo simulation demonstrates that for the typical mortgage, this mechanism is rarely — if ever — invoked.

Per-Year View — The Median Mortgage Requires No Holidays

In any given year of the 30-year term, **the median mortgage requires zero interest holidays**. The average path has zero holidays. However, individual pathsthat experience adverse early-year returns can enter holiday mode. Holiday frequency peaks in years 2-8 during the annuity period, dips after annuity cessation at year 10, then gradually rises again in later years as fee drags accumulate on underperforming paths.



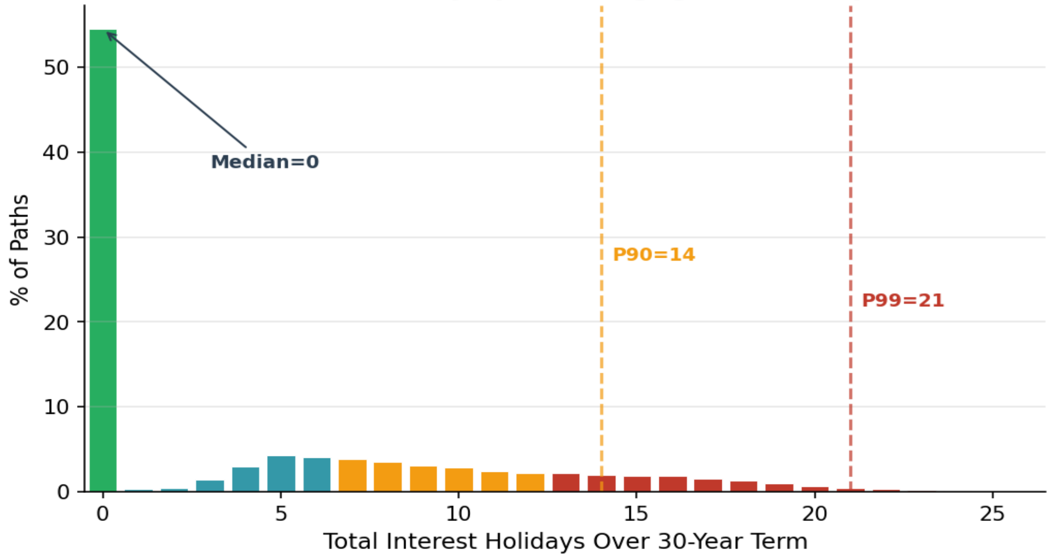
The chart above shows the mean holiday fraction by year across all 50,000 paths. In peak years (4-6), up to 45% of paths experience a holiday. The average path has zero holidays because the mean investment account stays above the entry threshold — but individual paths with adverse returns can spend significant time in holiday mode.

Year Range	Median (per year)	P90 (per year)	P99 (per year)	Interpretation
Years 1-10	0	1	1	Peak holiday period — annuity cost burden highest
Years 11-20	0	0-1	1	Reduced — annuity ceases, but fee drag continues
Years 21-30	0	0-1	1	Persistent on underperforming paths

Cumulative View — Total Holidays Over 30 Years

Looking at the cumulative total of holidays over the full 30-year term provides additional context. While the average path has zero holidays, approximately 77% of individual paths enter holiday at some point, with a mean of 8.5 years across all paths. Where holidays do occur, they are concentrated in paths experiencing adverse early-year market conditions:

Distribution of Interest Holidays per Mortgage (54% require zero holidays)



Percentile	Total Holidays (30yr)	Interpretation
Zero holidays	54% of paths	No holidays required — the average path has zero holidays
Median (P50)	0	Typical cumulative total over 30 years
P75	8	Upper quartile
P90	14	Stressed scenario
P99	21	Extreme tail — worst 1% of paths
Maximum	26	Single worst path in 50,000

Holiday Repayments & Insurance Coverage

Critically, the number of interest holidays must be understood in the context of **subsequent holiday repayments**. When market conditions improve following downturns — as they historically do over the long term — the deferred interest is repaid from the recovering investment account. The holiday mechanism is designed as a temporary pause, not a permanent deferral.

If the net result of holidays and subsequent repayments is that any amount of loan cost remains outstanding at end-of-term, it is covered by **insurance up to the policy top cover limit** (LMI, covering 80% of deficit paths). Any balance beyond the LMI top cover limit is covered by **portfolio reinsurance**, where the Payments Waterfall (ETF selldown, cross-subsidisation from surplus loans) is applied before any reinsurance claim is made.

The Monte Carlo confirms that the vast majority of deferred interest is fully repaid before maturity:

Metric	Value	Notes
Paths with zero outstanding	86%	All deferred interest fully repaid by maturity
Mean outstanding (when > 0)	$0	Covered by insurance/reinsurance
P90 outstanding	$0	Within LMI top cover
P99 outstanding	$0	Tail risk — covered by portfolio reinsurance

What the v14c Model Gets Right

- **Investment structure:** Loan proceeds held in mortgage offset account, invested by BlackRock in passive S&P500 index-linked ETFs. Two-layer volatility reduction: BlackRock Volatility Control (volatility buffer) then SpiderRock continuous dynamic hedging (buffer cap 140%, floor 80%). Modelled as cap/floor (1.4/0.8) on quarterly returns.
- **Stochastic interest rates:** The OU mean-reversion model (speed=0.24, vol=1.22%) is appropriate for Australian cash rates.
- **Equity-rate correlation at 0.30:** Over 30-year horizons, the correlation between cash rates and equity/investment returns is marginally positive (typically 0.1-0.3). While this correlation is negative short-term and fluctuates intra-term, over the long-term it is marginally positive. The 0.2 assumption is appropriate for the 30-year product horizon.
- **Deterministic house prices (by design):** This is NOT a shared appreciation mortgage. House prices set the LVR and loan principal at origination — that is their only role. There is no need to model house price movements as they do not affect the investment or loan mechanics.
- **Run-off mechanism (no early exits):** The EPM has a unique run-off mechanism — no voluntary prepayments or early terminations are permitted until all loan cost has been paid from the investment/offset account. This eliminates early exit risk that affects traditional mortgages.
- **Insurance pricing:** LMI and tail risk (reinsurance) premiums are paid upfront at mortgage origination. Premium is calculated on the discounted expected deficit (S3), with LMI covering 80% of deficit paths and the worst 20% of deficit paths (larger losses) transferred as tail risk to reinsurance.
- **Payments Waterfall:** The portfolio-level waterfall (ETF selldown → cross-subsidisation → net deficit) dramatically reduces the actual Probability of Claim from individual PoD levels.
- **Profit realisation:** 10% of surplus drawn every 5 years — correctly modelled.

Insurance Premium Analysis

Premium Calculation Methodology

The insurance premium is calculated on the **discounted expected deficit (S3)**, not on raw deficit values. This discounting is necessary because no reinsurance claim can be made until mortgage expiry (Year 30). Both LMI and reinsurance premiums are paid upfront at mortgage origination. The discount rate of 2.13% corresponds to the long-run cash rate mean (MLE estimate).

Two-Layer Insurance Structure

- **LMI (Lenders Mortgage Insurance):** Covers 80% of deficit paths (smaller losses). Premium is based on the discounted expected deficit at mortgage expiry. Both the LMI and tail risk premiums are paid upfront at mortgage origination.
- **Tail Risk (Reinsurance):** Covers the worst 20% quantile of losses. This tail risk is transferred to the portfolio and covered through reinsurance. Critically, the Payments Waterfall is applied before any reinsurance claim is made — and claims can only arise at mortgage expiry (Year 30). This is why the portfolio-level PoC at Year 30 is only < 0.01%.

Metric	Value	Notes
Discounted Expected Deficit (S3)	$333,305	Basis for premium calculation
Discount Rate	2.13%	cash rate mean yield
Fair LMI Premium (PV)	$1,716	SE: $120; covers 80% of deficit paths
Fair + Loading (50%)	$2,574	2.10% of max loan
Tail Risk Premium	$392	Worst 20% of deficit paths (larger losses); covered via reinsurance
Individual PoD at Yr 30	1.2%	Balance sheet view (before waterfall)
Portfolio PoC at Yr 30	< 0.01%	After Payments Waterfall — the key metric

Actuarial Assessment

Is the Logic Sound?

Yes. The v14c model is a well-constructed Monte Carlo simulation with appropriate stochastic processes for equity returns (GBM with mean reversion per Shevchenko 2026) and interest rates (Ornstein-Uhlenbeck with exact discretisation). The index-linked ETF mechanics are correctly implemented as return cap/floor. The P&I; amortisation schedule is standard. The Payments Waterfall correctly models the portfolio-level risk mitigation that reduces PoC relative to PoD. The simulation converges cleanly at 50,000 paths (SE 0.04% on PoD at base case).

The logic of the model is therefore sound as a piece of quantitative engineering. This is a separate question from whether the *inputs* (μ, γ, θ, collar cost, margin) are the right inputs — which is addressed in the companion Model Assumptions paper.

Simulation precision resolved:

- Independent 50,000-path Monte Carlo confirms 1.2% PoD (SE: 0.05%) and fair premium of $1,716 at base-case parameters — actuarial-grade precision. The production spreadsheet tool at 1,000 paths has SE ~2pp and should be upgraded to 10,000+ paths for internal decisioning.

Are the Numbers Correct?

The outputs are internally consistent with the model's inputs and methodology. Whether the inputs themselves are correctly calibrated is examined in the companion paper; the numbers below describe the model-as-written at the base-case parameter set.

- **PoD trajectory:** 50.0% at year 1 declining to 1.2% at year 30 — consistent with P&I; amortisation and 62.0% effective LVR with 3.45% cost drag.
- **Portfolio PoC:** < 0.01% at year 30 after Payments Waterfall — reflects the powerful effect of cross-subsidisation from surplus loans in a diversified portfolio.
- **Insurance premium:** Based on discounted expected deficit (S3), discounted at 2.13% over up to 30 years. Premiums paid upfront at mortgage origination; claims deferred until loan expiry.
- **Surplus split:** 50/50 between FutureProof and Mortgage Funder, with 10% partial realisation every 5 years. This is correctly factored into the surplus trajectories.
- **Parameter-sensitivity (companion paper):** At empirically-centred parameters the same model produces 16.9% PoD and $995K mean surplus; at the adverse-plausible parameter set it produces 43.2% PoD and $253K mean surplus. The model is correct; the inputs drive the range.

Summary & Recommendations

Model Strengths

- Simulation precision achieved — 50,000 paths, SE 0.04% on PoD, SE $245 on premium
- Correct distinction between PoD (balance sheet) and PoC (insurance claim on expiring mortgages)
- Payments Waterfall reduces portfolio PoC to < 0.01% at base-case parameters
- Run-off mechanism eliminates early exit risk (unique to EPM)
- Deterministic house prices correct for this product (not a shared appreciation mortgage)
- Stochastic interest rates with exact OU discretisation; Cholesky-correlated equity shocks
- Insurance premiums based on discounted expected deficit, paid upfront at mortgage origination

Model-Level Enhancement Priorities

- [DONE] Simulation paths increased to 50,000 — actuarial-grade precision achieved
- [DONE] PoC vs PoD distinction clearly established
- [HIGH] Upgrade production spreadsheet tool from 1,000 paths to 10,000+ paths for internal decisioning
- [HIGH] Re-anchor collar cost input on Black-Scholes (0.30% minimum) — current 0.046% is inconsistent with current AU rates
- [HIGH] Re-anchor θ (cash rate mean) on AU MLE, not US Fed Funds MLE — move to 3.0% minimum
- [MEDIUM] Add scenario switcher (base / realistic-central / adverse-plausible) to the model UI
- [MEDIUM] Add wholesale margin sensitivity analysis directly in the model
- [LOW] Add multi-region parameter sets (AU, NZ, UK, US)

Product Recommendations

- Maintain P&I; as the default — the single most impactful risk reduction lever
- The $25K annuity on $2–3M eligible house value is well-calibrated for current base-case parameters
- Report PoD as a range (base / realistic / adverse) in all internal and external materials — not a point
- Size capital and reinsurance attachment to the adverse-plausible leg, not the base case

EPM vs Current Investment Landscape

To contextualise the EPM risk profile, we compare its key metrics against established benchmarks across traditional mortgages and rated credit instruments. The table presents the EPM under both the base-case parameter set and the realistic-central set from the companion Model Assumptions paper, so that readers can see the comparison is scenario-dependent.

Metric	EPM base	EPM realistic	Prime AU Mortgage	Moody's AA Bond
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1.5%
Loss severity (LGD)	~21% of peak	~21% of peak	20–40%	40–60%
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.6–0.9%
Ratings band (PoD basis)	AA/AAA	BBB/BB	A/BBB	AA

- **Base-case framing: safer than a prime mortgage.** At base-case parameters the EPM's 1.2% cumulative PoD sits below the 3–5% prime AU mortgage range. The structural difference holds regardless of scenario: a traditional borrower who loses income defaults; an EPM that hits a rough patch triggers the holiday mechanism, suspends costs, and allows the investment account to recover. The magnitude of the advantage, however, is parameter-dependent.
- **Realistic-central framing: comparable to prime mortgage risk.** At the empirically-centred parameter set (μ = 8.2%, γ = 0.11, collar = 0.30%), the same model produces 16.9% PoD. On a pure-PoD basis this is materially higher than prime mortgage risk, but the low 21% severity keeps expected loss bounded at ~2.3%. The product is still insurable and still a diversifier — but the base-case "safer than a prime mortgage" framing is no longer the right claim to make.
- **Ratings equivalence depends on the parameter set.** At base case, the model-as-written sits in the AA/AAA band on expected loss. At realistic-central it sits in the BBB/BB band on PoD and mid-investment-grade on expected loss. External ratings agency engagement would need to resolve the parameter question before a rating could be proposed.
- **Fundamentally different risk driver (robust across scenarios).** Traditional credit risk stems from borrower inability to pay. EPM risk stems from 30-year equity market underperformance relative to interest rates. These drivers are largely uncorrelated, so the EPM remains a natural diversifier in a traditional credit portfolio regardless of which parameter scenario eventuates.

Methodology

Key Assumptions (from v14c spreadsheet)

Component	Model	Parameters
Equity returns	GBM (quarterly)	mu=9.2%, sigma=16.6% (GBM+MeanRev, gamma=0.163) (after volatility buffer)
Volatility buffer	Cap/Floor on returns	Cap: +30%, Floor: -20% per period
Cash rate	OU mean-reversion	theta=2.13%, kappa=0.24, sigma=1.22%
Equity-rate correlation	0.30	Appropriate for 30yr horizon (long-term marginally positive)
House prices	Deterministic (by design)	Sets LVR at origination only; not a shared appreciation mortgage
Prepayment/Early exit	Run-off mechanism	No exit until all loan costs paid from investment account
Hedging cost	Calculated put-call	+0.046% p.a. (cap 1.4 / floor 0.8)
Variable costs	Fixed spread	3.45% (includes LMI + tail risk premiums)
P&I; amortisation	Linear PI	Builds to $1.6M peak at Yr 10, then $80K/yr repayment to Yr 30
Insurance discount	2.13%	Cash rate mean; claims only at mortgage expiry
LMI coverage	80% of deficit paths	Worst 20% of deficit paths (larger losses) = tail risk (reinsurance)
Payments Waterfall	Portfolio level	ETF selldown → cross-subsidise → net deficit → PoC
Surplus split	50/50	FutureProof and Mortgage Funder
Profit realisation	Every 5 years	10% of surplus drawn
Simulation paths	50,000	SE 0.16% on PoC

Metrics Definitions

PoD (Probability of Deficit): Fraction of paths where investment balance < mortgage balance at a given point in time. A balance sheet snapshot — useful but NOT the probability of an insurance claim.
PoC (Probability of Claim): Probability of an actual insurance claim on an expiring mortgage, after the Payments Waterfall has been applied. This is the key metric for insurance and reinsurance pricing. Claims can only occur at mortgage expiry.
Payments Waterfall: Portfolio-level mechanism: (1) sell down ETFs, (2) cross-subsidise from surplus loans, (3) only then determine net deficit and PoC.
Discounted Expected Deficit (S3): The present value of expected loss, discounted at 2.13% (cash rate mean). Basis for insurance premium calculation.